NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Biometric Security — Course Project

Contactless-to-Contact Fingerprint Recognition

Cross-Modal Relational Alignment (CMRA) with Risk-Controlled Abstention

A reproducible study on the HKPolyU Contactless 2D to Contact-based 2D database

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Why contactless-to-contact is still hard

- Contactless capture removes the sensor, but not the legacy contact database.
- Cross-modal gap: illumination, pose, perspective, ridge scale and sensor noise change together.
- VerifNet-style systems still rely on a Siamese branch plus fixed 90/10 score fusion.
- The strongest 2025–26 methods need transformers, external data or synthetic augmentation.
- Biometric security is decided at low FAR, where calibration and abstention matter.

Hygiene + access

touchless capture, remote onboarding

Interoperability

match phone photos to legacy prints

Security

operate at FAR = 1% and 0.1%

Goal: one embedding that aligns cross-modal identity structure — not another fusion layer.

Data and evaluation protocol

336
fingers, 2 sessions

5,952
paired images (2 modalities)

101
held-out test identities

10,728
test pairs

- Every finger: 6 contactless + 6 contact impressions per session; 160 fingers return in session 2.
- Split by finger: 201 train / 34 validation / 101 test — no identity appears twice.
- Primary test: 10,728 pairs (5,364 genuine, 5,364 impostor); all genuine pairs are same-session.
- Supplementary cross-session test: 48 test fingers in both sessions, 3,456 genuine + 3,456 impostor pairs.
- Metrics: EER, TAR@FAR = 1% and 0.1%; 95% CI from 500 identity-cluster bootstrap resamples.

Model selection uses validation only; the test set is scored once after freezing.

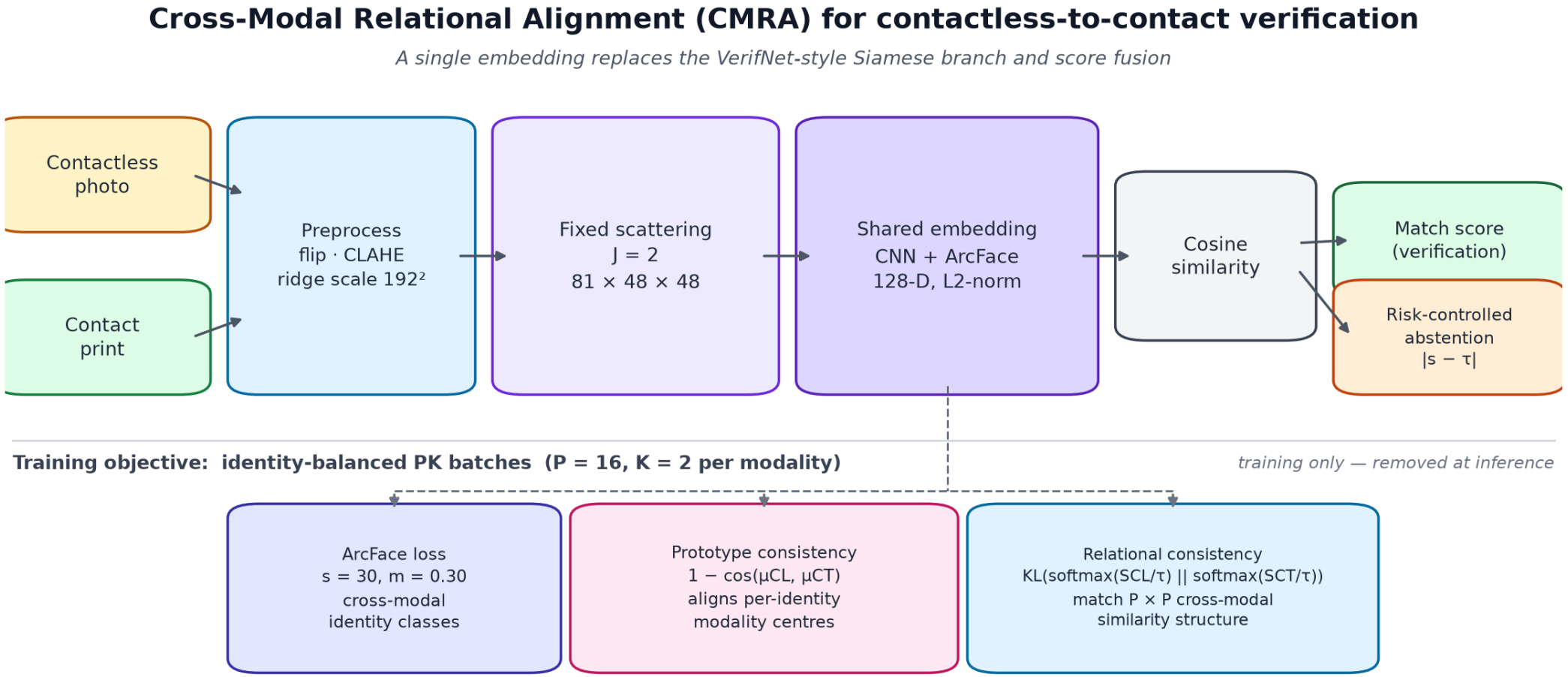
Related work: five papers, one gap

Paper	Core idea	Reported result
Lin & Kumar (TIP 2018)	Traditional ridge and minutiae features + COTS matcher	Contactless-to-contact baseline on HKPolyU
Tan & Kumar MANet (TIFS 2021)	Minutiae-guided attention CNN	Contactless-to-contactless identification
VerifNet (TBIOM 2025)	Scattering + Siamese + minutiae, fixed 90/10	2.5% EER, IITI-CFD; CL2CL only
Ridgeformer (ICIP 2025)	ViT + multi-stage contrastive, 4 training sets	2.83% EER, HKPolyU
Kaplesh et al. (EAAI 2025)	Systematic review of end-to-end pipelines	Domain shift and preprocessing remain open

- Common thread: representation, calibration and quality are entangled; fusion weights are usually fixed.
- Gap: no reproducible identity-disjoint CL2CB baseline built around a genuinely different training objective.

State of the art is 2–4% EER — but with external data, transformers or synthetic pipelines.

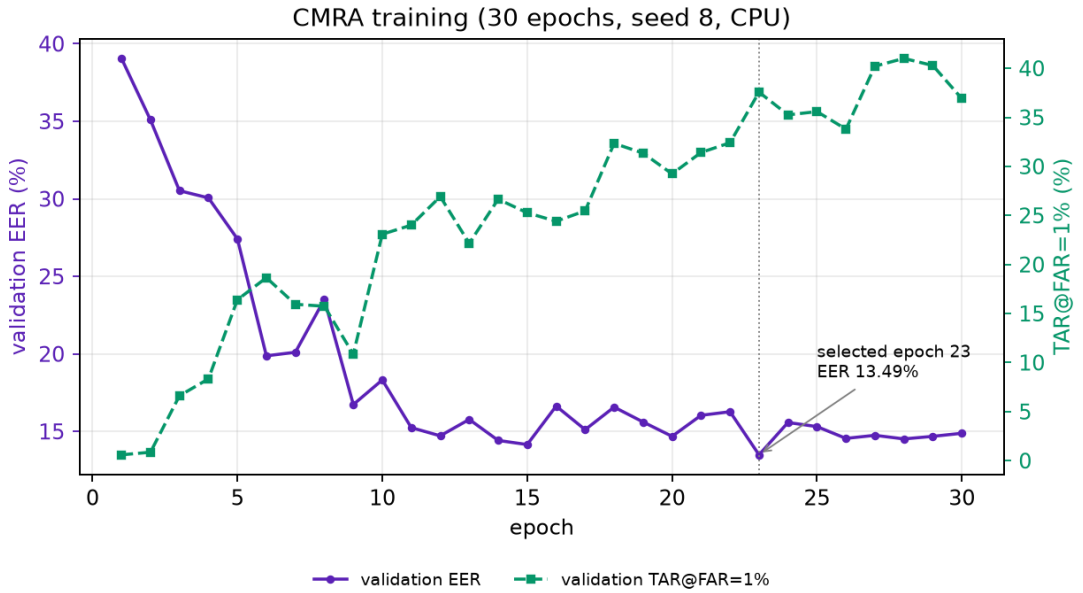
Proposed pipeline: CMRA



One shared embedding; the Siamese branch and score fusion are removed from the proposed system.

A CPU-cheap front-end, one shared embedding

- Fixed 2-D scattering transform (J = 2, 81 × 48 × 48) replaces learned early filters.
- Three conv blocks → 128-D L2-normalised embedding; about 0.13M parameters.
- ArcFace: scale 30, margin 0.30; both modalities share one identity class.
- 30 epochs, AdamW, ±2-cell translation, label smoothing 0.05.
- Checkpoint at epoch 23: validation EER 13.49%, test EER 16.06% (seed 8).



Trained from scratch on 201 identities — no external data.

Identity-balanced PK batches + ArcFace

P = 16
identities per batch

K = 2 + 2
contactless + contact per identity

64
images per training step

64
steps per epoch

- Sampled by identity, not by pair: every step contains both modalities of the same fingers.
- Random crop ±2 cells; the identity split is never violated.
- A shared class label lets ArcFace learn the cross-modal identity structure directly.

ArcFace supervision

$$L_{arc} = -\log\left(e^{s\cdot\cos(\theta_y + m)} / \Sigma_j e^{s\cdot\cos(\theta_j)} \right)$$

- s = 30, m = 0.30; label smoothing 0.05.
- θ_y is the angle between the embedding and its class weight.
- Contactless and contact images of the same finger share the label y.
- This replaces pairwise contrastive training with identity-level supervision.

Identity-level supervision, not pairwise contrastive loss.

Two cross-modal consistency losses

Prototype consistency

$$L_{proto} = (1/P) \Sigma_i [\, 1 - \cos(\mu_i^{CL}, \mu_i^{CT}) \,]$$

Pulls the mean embedding of each identity across the two modalities together.

Relational consistency

$$L_{rel} = \tfrac{1}{2} KL(p_{CT} \,\|\, p_{CL}) + \tfrac{1}{2} KL(p_{CL} \,\|\, p_{CT})$$

p = softmax(S / τ); aligns who-is-close-to-whom across modalities.

$$L = L_{arc} + \lambda_{proto} \cdot L_{proto} + \lambda_{rel} \cdot L_{rel} \quad \text{with } \lambda_{proto} = \lambda_{rel} = 0.5,\ \tau = 0.1$$

- Prototype loss aligns modality centres; relational loss aligns the full P × P similarity structure.
- Temperature τ = 0.1 keeps the KL term numerically meaningful — a raw Gram MSE was negligible (~0.001).
- At seed 7, relational-only (17.74% EER) is the strongest single addition, ahead of prototype-only (18.23%).

The network is trained on the cross-modal relation, not only on identity labels.

Ablation: where the gain comes from

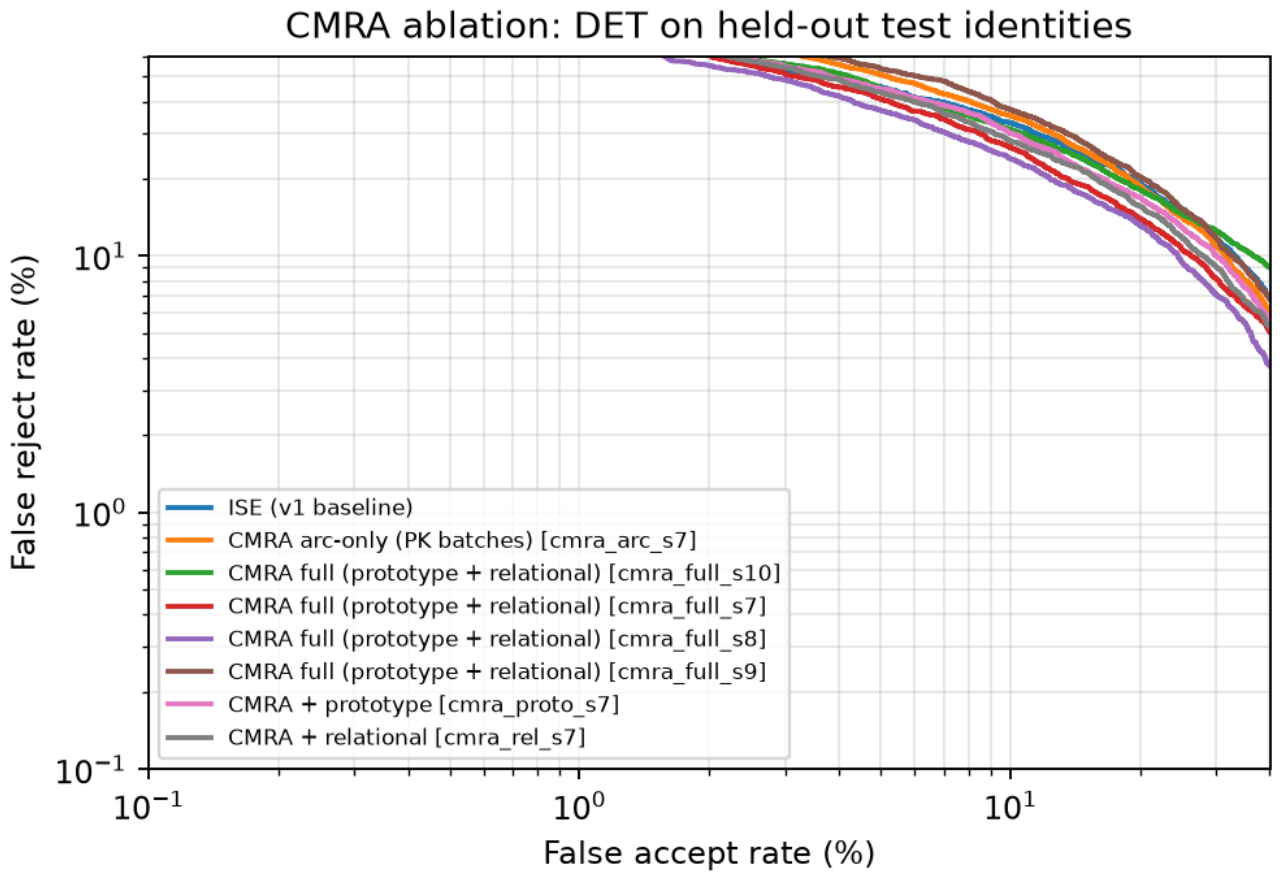
Method	EER	TAR@1%	TAR@0.1%
ISE (v1 single embedding)	19.56	28.24	4.05
CMRA ArcFace + PK only	19.41	24.33	4.34
CMRA + prototype	18.23	27.44	5.87
CMRA + relational	17.74	28.93	6.99
CMRA full (prototype + relational)	**16.80**	**29.55**	**7.20**

Percentages; 10,728 test pairs, 101 identities, seed 7.

- ArcFace + PK alone (19.41%) is already at the ISE level (19.56%).
- Each consistency term helps; full CMRA reaches 16.80% EER.
- v1 best fusion is 18.16% EER, but 32.94% TAR@1% — still stronger in the low-FAR tail.
- Four seeds span 16.06–20.15%; report mean ± spread, not one cherry-picked seed.

Single embedding, no score fusion — and a lower EER than the v1 fusion pipeline.

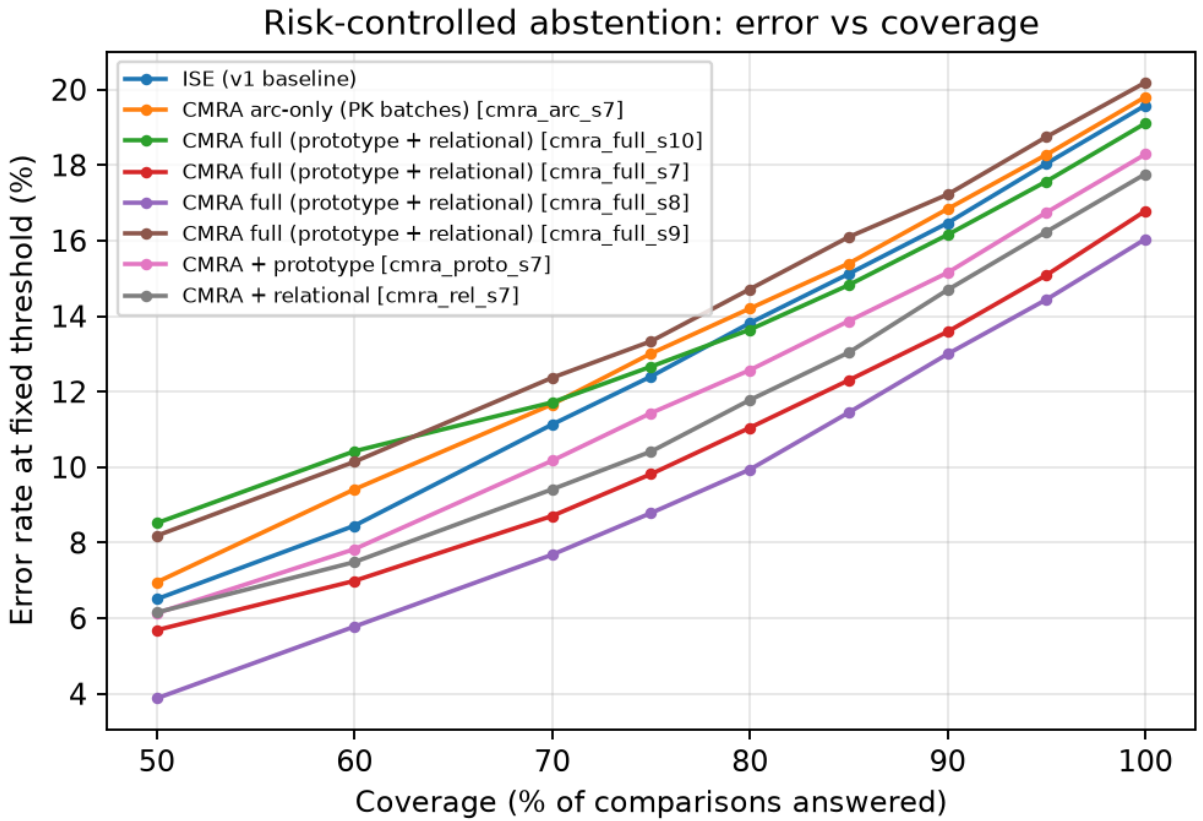
DET curves across the operating range



- CMRA full has the lowest EER of the ablation.
- ArcFace-only and ISE overlap; the consistency losses separate them.
- The low-FAR tail is close: v1 fusion keeps a small advantage at TAR@1% and 0.1%.
- DET uses log–log axes, so the tail is visible rather than hidden by the EER point.

Honest claim: comparable-or-better accuracy with a simpler, genuinely new pipeline.

Risk-controlled abstention



Coverage	Error rate
100%	16.04
90%	13.00
80%	9.94
70%	7.68
50%	**3.88**

- Rank by confidence $|s - \tau|$ and reject the least confident pairs.
- AURC 0.056 vs 0.084 for ISE.

At 50% coverage the error falls to 3.9% — a security-facing capability absent from v1 and the compared literature.

Does it survive a session gap?

Method	Cross EER	Cross TAR@1%	Same EER	Same TAR@1%
CMRA seed 8 (proposed)	**14.89**	**32.81**	**15.79**	**25.64**
v1 quality-aware fusion	18.50	27.66	16.26	30.35
ISE (v1 baseline)	18.89	31.57	17.43	28.36
Minutiae	37.66	9.90	37.72	6.86

- CMRA beats v1 fusion by 3.6 points cross-session.
- Cross-session EER is actually lower than same-session here.
- The gain is not a same-session artefact.

48 test identities present in both sessions; 3,456 genuine and 3,456 impostor cross-session pairs.

Conclusion

16.80% EER · 29.55% TAR@1% · 3.9% error at 50% coverage

What worked

- Cross-modal relational alignment.
- Single embedding, no score fusion.
- CPU-only: 30 epochs, 0.13M parameters.
- Honest identity-disjoint protocol.

Honest limits

- Seed variance: 16.06–20.15%.
- v1 fusion still better at TAR@1%/0.1%.
- Cross-session study covers 48 identities.
- No state-of-the-art claim.

Next

- Joint minutiae + identity learning.
- Learned pair-reliability model.
- Synthetic multi-pose data.
- OpenVINO / NPU deployment.

A genuinely different training objective — relational alignment — not another fusion of the VerifNet score.